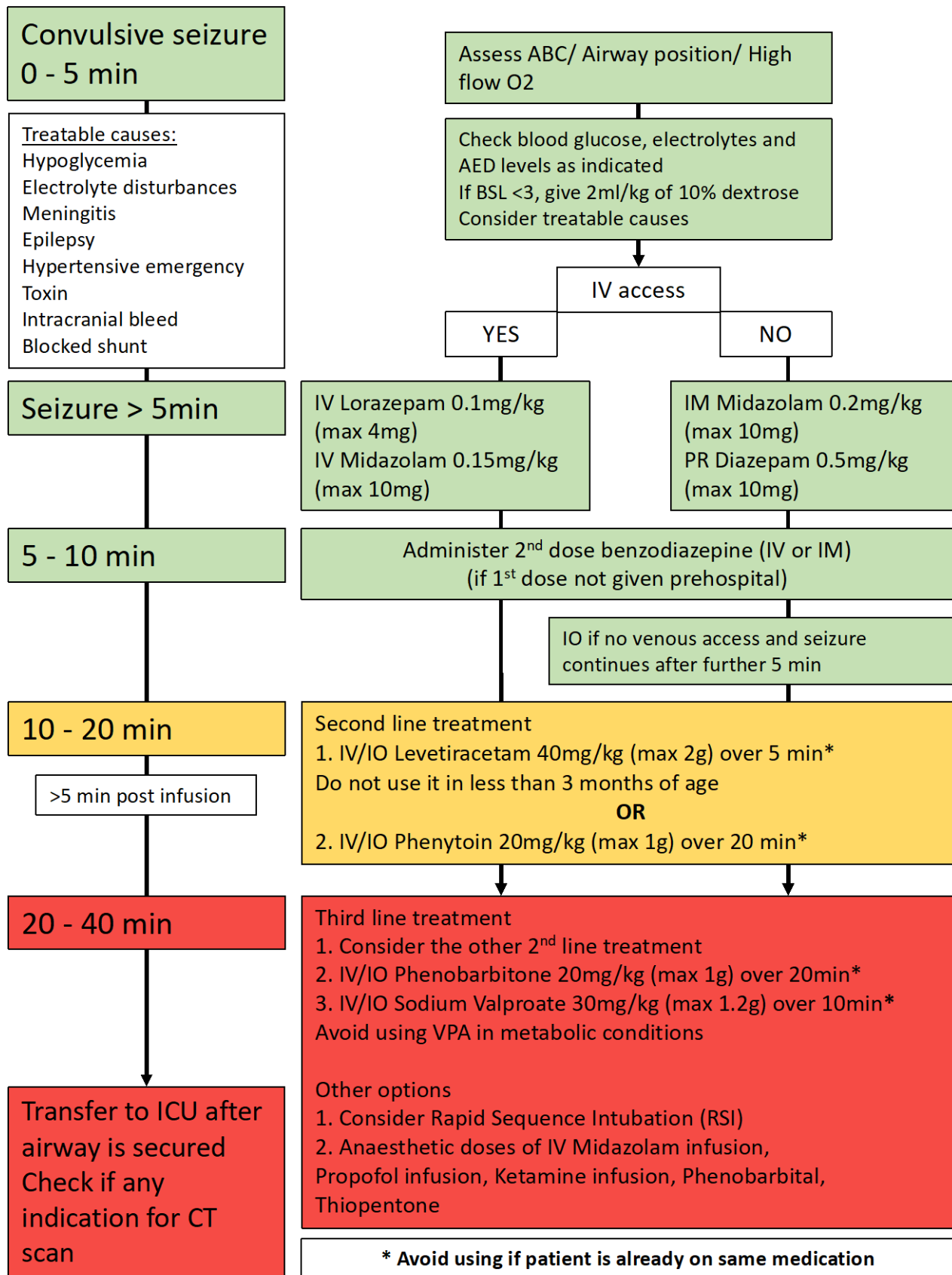


Paediatric Status Epilepticus Protocol (NOT FOR NEONATAL SEIZURE)



Guidelines for Status Epilepticus

Diagnosis

Principles in the management

Immediate management

Refractory status epilepticus

Management flow diagram

Diagnosis

Definition

SE is a condition resulting either from:

- the failure of the mechanisms responsible for seizure termination
OR
- from the initiation of mechanisms which lead to abnormally prolonged seizures (after time point t_1)

Type of SE	Operational dimension ¹ Time (t_1), when a seizure is likely to be prolonged leading to continuous seizure activity
Tonic clonic SE	5 min
Focal SE with impaired consciousness	10 min
Absence status epilepticus	10-15 min

Clinical course

- discrete overt seizures
- seizures waxing and waning in severity
- continuous convulsive SE
- subtle convulsive status
- comatose with electrical status

Principles in the Management

1. Support of vital functions
2. Controlling and aborting all seizure activity
3. Preventing or reversing physiological changes that may have taken place

Immediate Management

1. Ensure patient airway, breathing and circulation. Lie patient on one side and on a flat surface
2. Obtain IV access, and obtain relevant bloods including glucose, electrolytes and AED levels

Pitfalls in the treatment of Status Epilepticus

1. Inadequate emergency drug treatment (inadequate benzodiazepine doses e.g. $<0.3\text{mg/kg}$ /dose of diazepam)
2. Failure to give adequate maintenance anti-epileptic medication
3. Failure to identify or treat the underlying cause
4. Failure to treat other related medical complications, e.g. hypoxia, hypoglycaemic, electrolyte abnormalities, hypotension and hyperthermia
5. Misdiagnosis of pseudostatus

Failure of the above management may result in **REFRACTORY STATUS EPILEPTICUS**

Refractory Status Epilepticus

Defined as the persistence of status epilepticus despite appropriate and sufficient administration of at least 2 of the first line anti-epileptic drugs

Management of Refractory Status Epilepticus

1. Transfer patient to PICU and monitor body temperature, cardiac rhythm and oxygen saturation
2. Consider elective intubation of the patient if there is respiratory depression or if high dose phenobarbitone is to be used
3. Inform Neurology registrar/consultant immediately to consider therapeutic options
4. Options available include a continuous infusion of **midazolam** (0.2mg/kg bolus, then continuous infusion, usually needing increasing doses to maintain control)
5. Induce **barbiturate coma** with phenobarbitone 5-10mg/kg IV every 15-20 min until seizures are controlled. A minimum of 30 mg/kg of phenobarbitone is usually needed. There is **no** maximum dose. Monitor vital signs as pressors are occasionally needed, especially if doses of 40-60mg/kg are used. (Respiratory depression is more likely to be due to the seizure unless the phenobarbitone is administered too rapidly).
6. Alternatively, **thiopentone** can be used to induce barbiturate coma. Thiopentone causes more hypotension, but may be used by giving a 3-5 mg/kg bolus, then infusion at 1-5 mg/kg/hr
7. If EEG monitoring is available, doses can be titrated until a burst suppression pattern is achieved. The EEG should be checked several times a day for the burst suppression pattern. In the absence of EEG monitoring, the coma can be lifted every 48-72 hours to see whether the seizures have stopped.
8. The last resort is propofol or inhalational anaesthesia
9. Ensure adequate support, such as fluid restriction, hypertonic saline and hyperventilation to decrease cerebral edema. Keep glucose at least 5-12 mmol/L, and ensure that electrolytes and osmolality are not deranged. Check anti-convulsant levels every 12-24 hourly.

Reference

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3. S Shorvon, M Ferlisi. The Treatment of Super-refractory Status Epilepticus. *Brain* 2011;134(10):2802-2818.